Phase 2: Literature Review

Project: BloodLink | رابط الحياة
Program: DEPI (Digital Egypt Pioneers Initiative), 2026
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1. Background

- **Voluntary, unpaid donation is the safest model.** The World Health Organization (WHO) recommends that all blood comes from voluntary, unpaid donors. WHO's 2025 global report says these donors now give more than 85 % of the world's blood, but the share is much lower in low-income countries (63.4 %) [1].
- **Families are often asked to find donors themselves.** A 2017 review of blood transfusion in WHO's Eastern Mediterranean region names Egypt among the countries still "under the weight of family/replacement blood donation" [2]. In practice, the patient's family has to find donors quickly.
- **Social media fills the gap, but badly.** Families post "blood type X needed urgently" on Facebook and WhatsApp. These posts reach people by chance, can't be checked, stay online after the need is over, and publish phone numbers to everyone.
- **Egyptians are willing to donate.** For World Blood Donor Day 2026, a Ministry of Health campaign registered more than 15,000 donors in 12 hours and set a Guinness World Record [3]. The willingness exists. What's missing is a trusted way to connect a specific patient with a nearby donor.

2. Existing solutions

Solution	What it does	Strengths	Gaps
Facebook and WhatsApp posts	Families post urgent requests in groups and on their timelines	Huge reach, free, familiar	No checking, no matching by blood type or distance, phone numbers are public, posts stay up after the need ends
Facebook Blood Donations (in Egypt since June 2020, with the Ministry of Health) [4][5]	Users sign up as donors and get messages from the nearest blood transfusion center	Official partner, large user base	Connects donors to **centers**, not to a specific patient who needs blood now
HOPE / GiveMeHope (Egyptian app by the Hope foundation) [6][7]	Collects blood requests posted on Twitter, lets users post requests, and finds donors with the right blood type within 20 km. Earlier versions also worked by SMS.	Matching by blood type and distance, brings requests together in one place	Built on public posts, and requests aren't checked against hospital papers
Law 3andak Dam (2012) [8]	Matches donors and patients through Twitter and Facebook	One of the first Egyptian attempts	No way to verify donors or requests (noted by the source itself)
University projects (e.g. MSA blood donation app) [9]	Blood bank and donor management apps	Show the need is recognized	Usually built around blood banks, not a community with trust levels

3. The gap BloodLink fills

None of the solutions above combines all of the following. BloodLink does:

Need	How BloodLink answers it
Is the request real?	Every request needs an official hospital paper. Partner hospitals confirm it (**Verified**). Otherwise an AI checks it (**Paper looks valid**).
Is the donor real?	Donors upload an official blood-type certificate and get the same trust badges.
Will the right people see it?	Only compatible, available donors near the hospital are notified, in growing waves.
Is it safe?	Phone numbers stay hidden until a donor accepts, meetings happen only at the hospital, money is never allowed, and there's a report button.
Who is responsible medically?	Always the hospital. BloodLink only connects people.

4. Feedback & Evaluation

This section records the lecturer's assessment of the project. It is filled in after each review.

Date	Phase reviewed	Lecturer's feedback	Our action	Status
dd/mm/2026	Idea & planning			

Our self-evaluation (before the lecturer's review)

Area	Strength	Weakness we know about
Idea	Solves a real, common problem in Egypt	Real partner hospitals are not signed yet, so the demo uses fictional ones
Trust model	Clear badges that never over-promise	The AI can be fooled by a well-made fake paper
Scope	A fixed core with checkpoints and a freeze date	Six people on one codebase need strong coordination
Cost	Runs at $0 on free tiers for the demo	Free hosting sleeps after 15 minutes without traffic
Course fit	Uses the whole MERN course: React, TypeScript, Tailwind, Node, MongoDB, Docker, testing, prompt engineering	Push notifications and geo queries are new to the team

5. Suggested Improvements

Improvements we have already planned, and ideas for after the course. Suggestions from the lecturer will be added to this table.

#	Improvement	Why	When
1	English version of the interface	Reach foreigners living in Egypt and the evaluators	Sprint 5 (planned extra)
2	Map view of nearby requests	Easier to judge distance than a list	Sprint 5 (planned extra)
3	WhatsApp share card for a request	People already use WhatsApp; the card links back to the checked request	Sprint 5 (planned extra)
4	Donation history and simple charts	Motivates donors, shows impact	Sprint 5 (planned extra)
5	Signing real partner hospitals	More requests and donors get the stronger "Verified" badge	After the course
6	SMS alerts for donors without smartphones	Reach more donors	After the course (paid service)
7	Blood stock levels at partner hospitals	Donors can help before an emergency	After the course
8	Mobile app (Android and iPhone), built on the same backend	More reliable notifications, especially on iPhone	Future, not part of this project

6. Final Grading Criteria

DEPI grades the project on documentation, implementation, testing and presentation. We will fill in the exact marks once the lecturer announces them.

Area	What is graded	Marks
Documentation	Phases 1 to 4 and the final reports	*TBA*
Implementation	Source code quality, version control, README, deployed website	*TBA*
Testing	Test plan, automated tests, bug reports	*TBA*
Presentation	Slides, live demo, video, infographic	*TBA*
Total		*TBA*

7. References

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